Detection and Analysis of AI-Generated Social Media Content

Introduction

This project focused on exploring whether AI-generated text is distinct and distinguishable in comparison to human-written text within the context of social media. As large language models rapidly develop and become increasingly part of everyday communications, the line between AI and human authorship has become less clear. While prior work has explored AI text detection, there has yet to be much exploration of whether linguistic signals remain detectable in a social media context where writing styles are brief and highly contextual. Building on prior work[1] we extend the analysis specifically to short-form Twitter (currently known as X) and long-form Reddit texts, investigating whether AI-generated social media content shows detectable, differentiating patterns and leveraging these findings to train a model capable of distinguishing between AI and human posts across social media platforms.

We first explored unsupervised methods to investigate whether natural clustering could separate AI and human text without labels. We engineered and combined features from two perspectives, stylistic and sentiment features as well as semantic and contextual features utilizing SBERT embeddings. After reducing the feature space with Principal Component Analysis and applying K-Means to the resulting dimensions, we then measured clustering performance based on the Silhouette Score.

Then we implemented three models — Logistic Regression, Random Forest, and Support Vector Machine — trained on a balanced dataset of human-authored and AI-generated texts, utilizing stylometric, syntactic, sentiment, and structural features. Our Random Forest achieved the best performance, with 96.52% accuracy and an F-1 score of 0.9643 ± 0.0009. Feature importance analysis revealed that sentiment variance features such as subjective variance (13.7%) and polarity variance (10.12%) were the strongest predictors, confirming our emotional engineering hypothesis that AI-generated social media text maintains a more consistent emotional tone in comparison to human-authored text. Additionally, high precision (98.83%) indicates our model effectively minimizes false positives, successfully avoiding incorrect labeling of human-written content as AI-generated.

Related Work

We referenced the following studies during the course of this project:

1. The first reference is a similar project, “Hybrid Feature-Embedding Models for Robust AI Text Detection”.[2] The project utilizes an ai_text_detection_pile dataset from Hugging Face that includes long-form essays from Reddit and forums as well as GPT-generated texts, introducing linguistic features to distinguish between AI-generated and human-written text. Our project builds upon this as it explores features in specifically social media contexts to see whether we can identify which features are meaningful and whether this makes for a more robust detection in this specific domain.

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[1] Thomas, K., Knudsen, G., & Hardmeier, C. (n.d.). Hybrid Feature-Embedding Models for Robust AI Text Detection [The ACL Anthology]. https://aclanthology.org/2025.konvens-1.27.pdf.

[2] Thomas, K., Knudsen, G., & Hardmeier, C. (n.d.). Hybrid Feature-Embedding Models for Robust AI Text Detection [The ACL Anthology]. https://aclanthology.org/2025.konvens-1.27.pdf.

2. The second reference is the study, “Machines in the Crowd? Measuring the Footprint of Machine-Generated Text on Reddit”.[3] This project focuses on the prevalence and behavioral patterns of AI-generated text on Reddit specifically, as well as analyzing user engagement with these posts. Our project differs, as it does not only focus on one platform, rather identifying patterns and features within each platform and seeing whether this helps the model generalize across platforms.
3. The final reference is the study, “Ability of AI detection tools and humans to accurately identify different forms of AI-generated written content”.[4] This project compares scores from AI detection tools and evaluates them against human assessments. While the core idea of detecting AI-generated text is the same, the authors focus on detection in a healthcare publishing context rather than developing a classification model.

Data Sources

Our data sets for both the unsupervised and supervised learning aspects of this project were accessed from the following sources:

1. Twitter: https://github.com/kinit-sk/multisocial/blob/main/dataset/
This source provides data for multiple social platforms, so here we focus primarily on more public platforms such as Twitter (currently known as X).
This link includes two Twitter datasets.

- One Twitter dataset is 1.6 million rows and includes the tweet ID, date, user, and text content, as well as tweet polarity for tweets in 2009. The estimated size is 85MB and is in CSV format.
- The other Twitter dataset is separated into four files (check-worthy, claim, harmful, and attention-worthy) of approximately 2,900, 4,945, 4,543, and 4,539 rows and includes the content topic, tweed ID, URL, and text content of tweets from 2020-2021. The estimated size is 1.4MB and is in CSV format.

2. Reddit: https://huggingface.co/datasets/fddemarco/pushshift-reddit/tree/main
This Reddit dataset is separated into 220 files by post date (2012-2018) and includes the user, date, whether there were comments on the post, the score (known on Reddit as “karma”) of the post, the subreddit it was posted to, the post title, and the post content. It has 121,782,217 rows, its estimated size is 89.1GB, and it is in CSV format.. Due to its size, we significantly reduced the dataset to select subreddits (topic categories).

3. Synthetic (AI-generated)
This dataset was developed through prompting, using three similar prompts to the reference “Hybrid Feature-Embedding Models for Robust AI Text Detection” project above, and utilizes the large language model GPT-5.2 to ensure diversity and variability in the text data. The rows were created to reflect the human-authored data and features, including tweet or post text, word count, sentence count, and character count. Common AI responses and other artifacts that could act as data leakage were removed post-generation to control for quality. The content of the dataset was validated by confirming tweet and post text variance and correct character limits.

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[3] La Cava, L., Aiello, L. M., & Tagarelli, A. (2025). Machines in the Crowd? Measuring the Footprint of Machine-Generated Text on Reddit. https://arxiv.org/pdf/2510.07226.
[4] Cheng, A., Lin, Y., Reedy, G. *et al.* Ability of AI detection tools and humans to accurately identify different forms of AI-generated written content. *Adv Simul* 10, 66 (2025). https://doi.org/10.1186/s41077-025-00396-6

Feature engineering

Our feature engineering pipeline transformed raw social media text into a comprehensive set of 24 features capturing stylometric, syntactic, sentiment, and structural characteristics of writing (full list found under Part A Method). Raw text from Reddit Pushshift dataset and Twitter underwent multi-stage cleaning to handle noisy social media data. We applied normalization, removed URLs, hashtags, and handles, as well as performed NFKD unicode normalization to standardize character encoding. Non-standard characters were filtered while preserving essential punctuation, and multiple white spaces were collapsed. This preprocessing focused the models on writing style characteristics rather than content-specific artifacts.

Unsupervised Learning

Method
The unsupervised learning pipeline for this project consists of data loading, comprehensive feature extraction, feature scaling, embedding concatenation, dimensionality reduction, and clustering. To fully capture the multifaceted nature of the text data, we engineered and combined features from two distinct perspectives:

1. **Stylistic and Sentiment Features:** To reflect the subtle differences in writing style and emotional tone between human and AI authors on social media, we extracted an extensive set of nearly 40 stylistic and sentiment-based features. These include metrics such as sentiment polarity, subjectivity, burstiness, word count, vocabulary richness (Type-Token Ratio), and specific "rule deviations" (eg. consecutive punctuation frequency, emoticon usage, and all-caps ratio).

2. **Semantic and Contextual Features:** To capture the deep contextual meaning and nuances of the sentences beyond simple word frequencies, we utilized Sentence-BERT (SBERT) embeddings using the SentenceTransformer('all-MiniLM-L6-v2') model.

Before combining these two feature sets, StandardScaler was applied to the stylistic and sentiment features. This normalization ensures that the dense, high-dimensional SBERT embeddings do not disproportionately overshadow the variance of the stylistic metrics during distance calculations. This composite feature representation was chosen because distinguishing highly capable AI from human text requires an understanding of both what is being said (semantics) and how it is being expressed (stylometrics and emotion).

We applied the Dimensionality Reduction and K-Means Clustering unsupervised learning technique to explore and structure the text data. The high-dimensional concatenated feature space (SBERT + Stylistic/Sentiment features) was first reduced to two dimensions (n_components=2) using Principal Component Analysis (PCA). Subsequently, the K-Means algorithm (n_clusters=2) was applied to the reduced embeddings. K-Means operates on Euclidean distance, making it highly effective at grouping texts that are stylistically and semantically proximal into structural clusters. We chose this method to determine whether a distance-based algorithm could naturally partition the dataset into distinct "Human" and "AI" buckets without relying on ground truth labels.

Results

1. Choice of evaluation method
   To evaluate the performance of clustering model without relying on ground-truth labels, following metrics were utilized:
   a. Silhouette Score : Measurement on how similar an object is to its own cluster compared to other clusters. A score of **0.5289** was achieved, the best among other settings, indicating that the clusters are comparably well-separated and have a reasonably solid structure.
   b. Cross-Tabulation (Label Alignment): By comparing the unsupervised clusters (Cluster 0, 1) with the original labels (AI, Human), verification on whether the clusters naturally correspond to the authorship of the text was conducted.

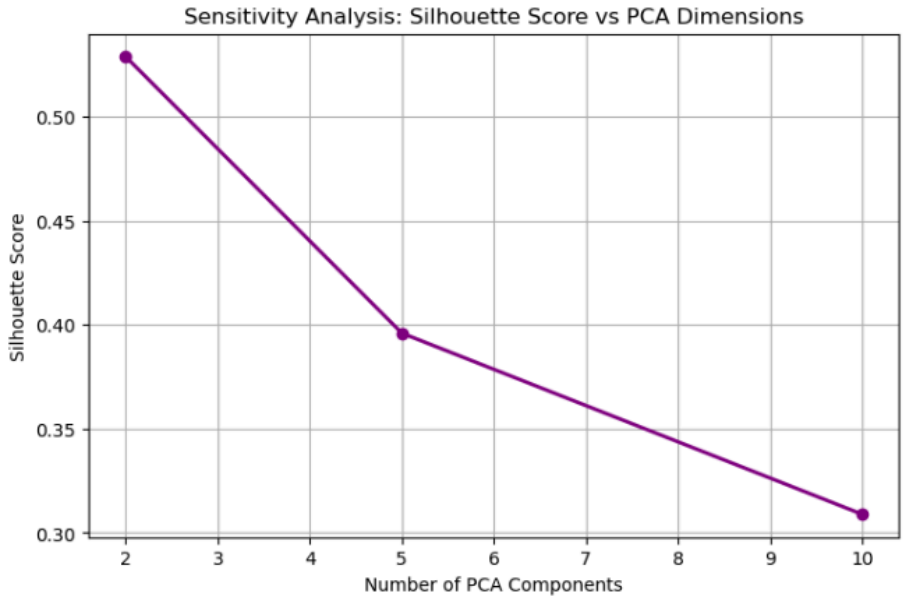
2. Overall summary of results
   a. **K-Means Clustering -** comparison between the best model vs other model
      The best-performing model utilized a **combination of SBERT embeddings and 40+ Stylistic Features**, reduced to two dimensions via **PCA**, and clustered using **K-Means**.

Settings	Silhouette Score	Cross Tabulation				Note
		Cluster 0 - Label: AI	Cluster 0 - Label: Human	Cluster 1 - Label: AI	Cluster 1 - Label: Human	
TF-IDF + PCA(Stylistic only)	0.2148	8.32% (9981)	0.05% (62)	41.68 % (50019)	49.95% (59938)	Too Low Silhouette Score → Need Improvement
TF-IDF + PCA(Stylistic + Semantic(TF-IDF)	0.9342	48.62% (58344)	50% (6000 0)	1.38% (1656)	0% (0)	Cross Tabulation invalid → K-means splitted well but not in AI vs Human
SBERT + PCA(Stylistic + Semantic(SBER T)	0.5289	21.94% (26328)	1.20% (1436)	28.06 % (33672)	48.80% (58564)	Best Model - Chosen

Cluster 0: AI Focused group
Cluster 1: Human focused Group

   b. **K-means Clustering - Sensitivity Analysis**
      To determine how the number of PCA Components(Hyperparameter) affects the clustering quality(result), the sensitivity analysis was performed.
      i. **Method:** Varied the number of PCA components (2, 5, 10) while keeping the K-Means parameters constant (k=2).
         (The number of k in K-means remained in 2 since the purpose of this

project is to determine characteristics between human and AI, which are two separate entities.)

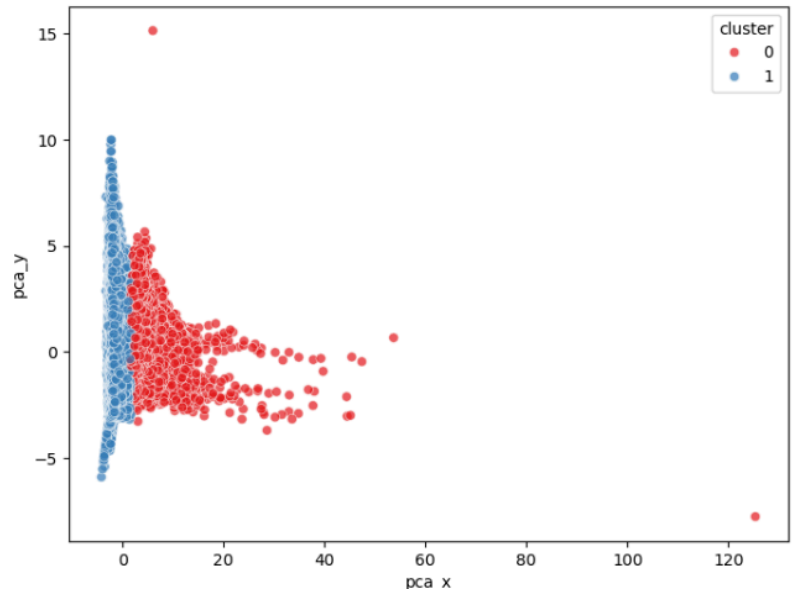


ii. **Findings:** Highest Silhouette Score observed at components=2.

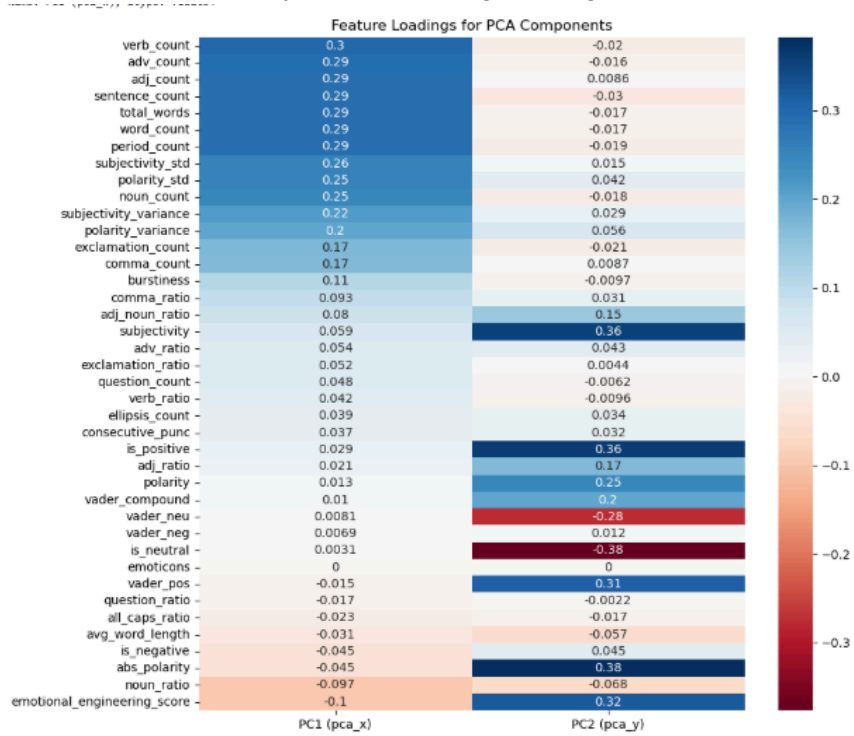
1. As the number of dimensions increased, the Silhouette Score decreased.
2. This can be interpreted as the "Curse of Dimensionality" (the increasing dimensionality introduces noise or "spreads out" the data in a way that makes the cluster boundaries less distinct).
3. [Conclusion] Reducing the data to 2 primary components is the optimal strategy for this dataset (it retains the most discriminative stylistic information while maintaining high cluster density.)

c. K-means clustering - Findings / visualization

i. Cluster 0 and 1 is well separated visually, with PCA x(PCA 1) on the x axis and PCA y(PCA 2) on y axis

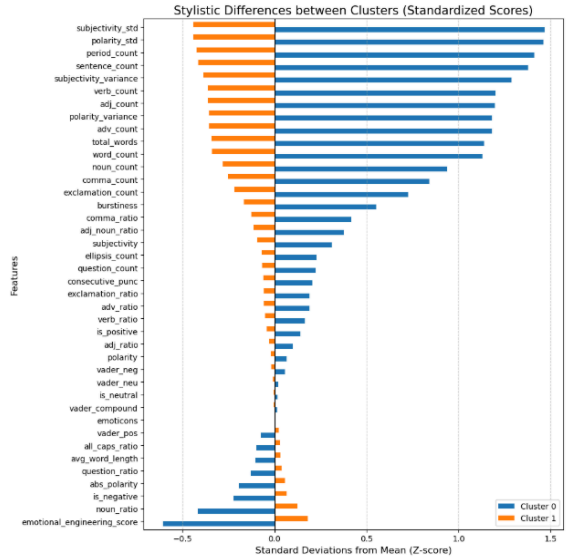


ii. PCA1 focuses on stylistic characteristics
(ex. verb/adv/adj/sentence count, comma counts)
PCA2 focuses on semantics characteristics
(ex. Is_positive, polarity, emotional engineering score)



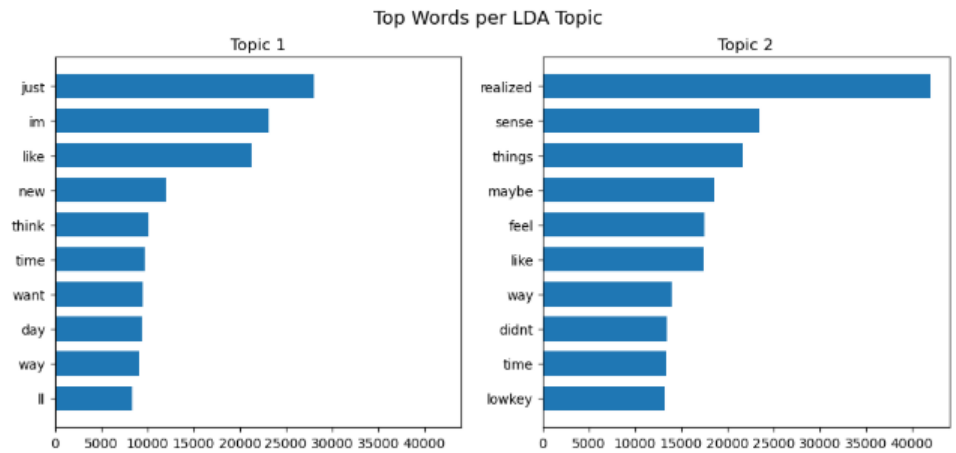
iii. Cluster 0(AI focused group) shows high vocabulary complexity(ex. avg_word_length, noun ratio) and high emotional variance(polarity_variance, subjectivity_std). Cluster 1 (human focused group) shows high on emotional, engineering score, noun ratio, is

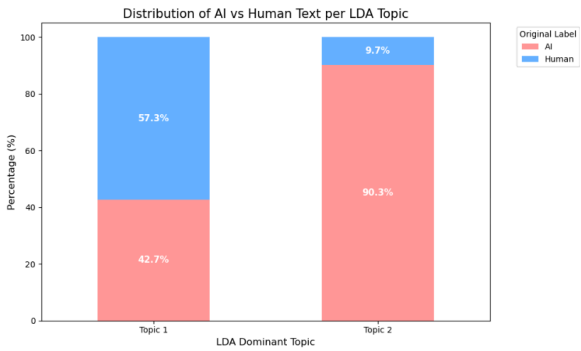
negative

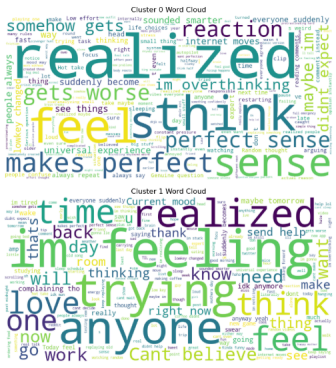


d. LDA - Findings / visualization

i. Based on the distribution of Topic 1&2 from LDA, 90.3% was AI on Topic 2.







Supervised Learning

Method

The supervised learning workflow consisted of collecting labeled samples, through the concatenation of human-authored and AI-generated datasets, preprocessing the text, and implementing the three following models:

1. Logistic Regression: To establish a linear baseline for feature importance. Tuned hyperparameters included regularization and penalty type to control overfitting.

2. Random Forest: To identify non-linear relationships between stylistic features. Tuned hyperparameters included the number of trees and feature sampling to balance variance and bias.

3. Support Vector Machine (SVM): To find the optimal decision boundary in a high-dimensional feature space. Tuned hyperparameters included regularization to control overfitting.

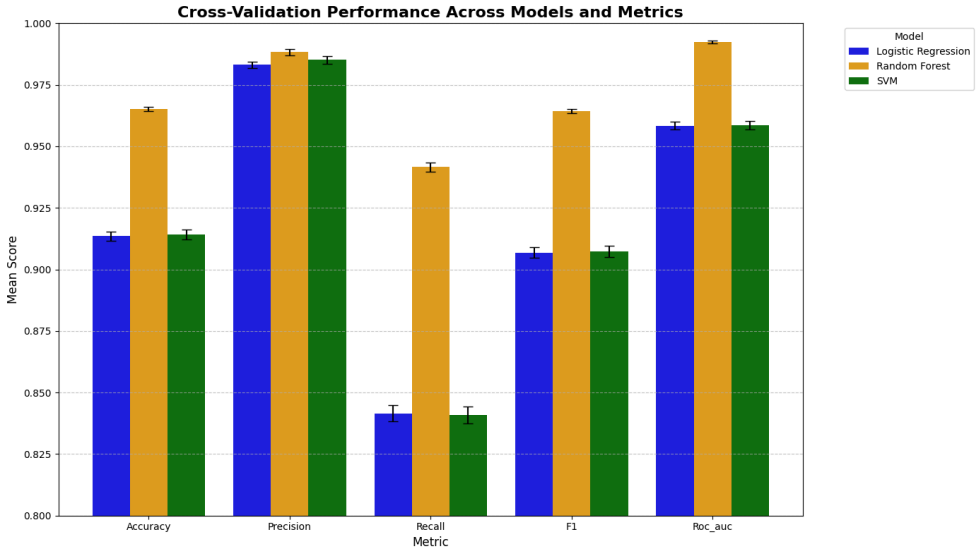
The feature representations used for the models include:

- Stylometrics: Word/sentence counts, punctuation patterns, and typo frequency (higher in human data).
- Syntactic Complexity: POS tags (via SpaCy) to calculate the ratio of descriptive adjectives to functional nouns.
- Strategic Sentiment: Unsupervised scores (polarity/subjectivity) to detect the "emotional engineering" common in AI scams.
- Structural Metrics: "Burstiness" (variance in sentence length) and T-unit density to measure the structural consistency of AI versus the erratic nature of human writing.

Evaluation

Models were trained, then evaluated with metrics such as Average Precision, Recall, and F1-score.

- Accuracy: To measure the overall proportion of correct predictions.
- Precision: To indicate our model effectively minimizes false positives, or successfully avoids the incorrect labeling of human-authored content as AI-generated.
- Recall: To measure all positively identified predictions.
- F1: To find a balance between the precision and recall evaluation scores.
- AUC-ROC: To indicate how well the model separated classes.



	Logistic Regression	Random Forest	SVM
Accuracy	0.9136 (±0.0019)	0.9652 (±0.0009)	0.9141 (±0.0020)
Precision	0.9831 (±0.0013)	0.9883 (±0.0013)	0.9851 (±0.0015)
Recall	0.8416 (±0.0034)	0.9416 (±0.0019)	0.8408 (±0.0036)
F1	0.9068 (±0.0022)	0.9643 (±0.0009)	0.9073 (±0.0023)
AUC-ROC	0.9584 (±0.0016)	0.9923 (±0.0006)	0.9585 (±0.0017)

*Mean across 10 cross-validation folds.

The resulting measurements revealed that Random Forest was the best performing model.

To identify which features drove the Random Forest model's detection capability, we conducted comprehensive feature importance analysis and systematic ablation testing.

Feature Importance Analysis

The Random Forest model's native feature importance scores reveal that sentiment variance features dominate the decision-making process. The top features, subjectivity_variance (13.7%) and polarity_variance (10.1%), collectively account for nearly 24% of the model's predictive power, validating our score hypothesis that AI-generated text exhibits more consistent emotional tone compared to the natural burstiness of human writing. Basic stylometric features like char_count (7.7%) and word_count (5.5%) provide moderate but consistent signals, while part-of-speech (POS) features, adv_count (9.2%) and verb_count (3.9%), capture differences in the grammatical structure between human-authored and AI-generated text.

Ablation Analysis

Through ablation analysis, punctuation features proved most critical — removing them caused a sharp 4.85% decrease in the F1-score. This indicates that emotional expressiveness patterns

are essential to discriminate signals, and that they cannot be compensated with other features. Sentiment features also showed moderate importance with a 3.39% drop in F1-score when removed. While sentiment features are important, the model partially compensates with punctuation and stylometric features, suggesting that there is some redundancy in capturing emotional signals. The removal of POS and stylometric features, on the other hand, caused only a 0.69% and 0.54% loss in the F1-score respectively, indicating syntactic complexity is largely captured through other features and that basic length statistics are redundant with more sophisticated features.

Isolation Testing

Additionally, isolation tests were conducted to determine how well feature groups perform alone. Stylometric features alone achieved 91.77% of baseline performance (F1: 0.8850), demonstrating strong standalone predictive power, and sentiment features alone reached 90.12% (F1: 0.8690), confirming their individual efficacy. However, punctuation and POS features alone dropped to 85.4% and 85.35% respectively, indicating they require complementary features for optimal performance.

Sensitivity Analysis

To assess robustness and identify potential overfitting or underfitting issues across hyperparameter tuning, a sensitivity analysis was conducted on the Random Forest model. We see that performance quickly stabilized as the ensemble size increased.

Number of trees (n_estimators)	F1
10	0.9567 (±0.0336)
50	0.9594 (±0.0313)
100	0.9595 (±0.0312)
200	0.9598 (±0.0310)
500	0.9598 (±0.0310)

The model exhibits low sensitivity to tree count beyond 100 estimators, with performance gains plateauing at ±0.0003 F1 (0.03%) between 100 and 500 trees. This indicates the ensemble converges quickly, suggesting our features provide strong, consistent signals that don't require extensive model averaging. Diminishing returns beyond 100 trees validate our choice of this value for deployment, as adding 400 more trees simply increases training time five-fold in exchange for negligible improvement in accuracy.

Tree depth (max_depth)	F1
5	0.9025 (±0.0347)
10	0.9275 (±0.0296)
20	0.9579 (±0.0334)
None	0.9595 (±0.0312)

In the above table, model performance shows strong tree depth dependency. Shallow trees, defined as a depth ≤ 10, severely underfit the data and cause a 3-6% loss in the F1 score, indicating AI detection patterns require complex decision boundaries with multiple feature interactions. The sharp increase from depth=10 to depth=20 (+3% F1) suggests critical stylometric patterns emerge from deeper feature combinations — likely from interactions between sentiment variance, punctuation, and POS features. Notably, unlimited depth, or None, does not overfit despite deep trees, as evidenced by consistent cross-validation scores (±0.0312 std). This indicates genuine pattern learning rather than memorization, likely due to Random Forest's built-in regularization.

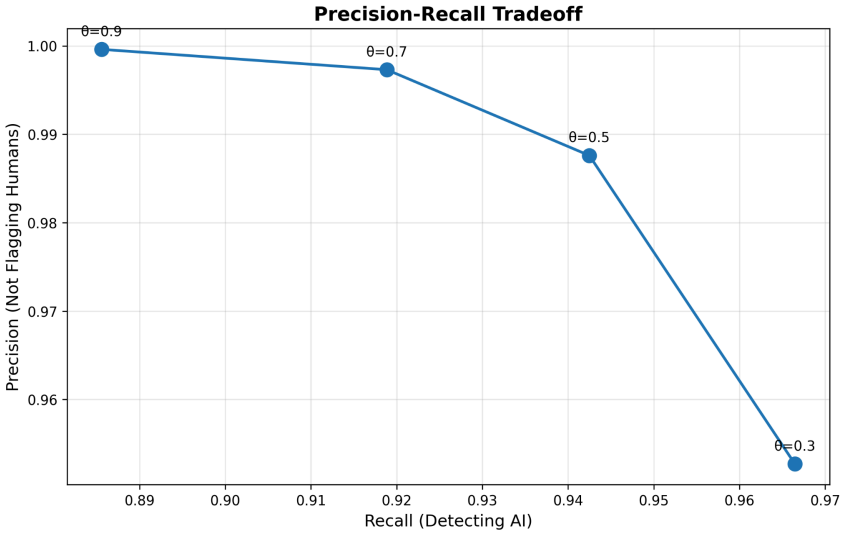
Tradeoff Analysis

The above model evaluation revealed three critical tradeoffs that can inform deployment decisions and also highlight inherent tensions in AI text detection systems.

1. Precision vs. Recall

Adjusting the classification decision threshold exposes a fundamental tradeoff between minimizing false predictions (precision) and detecting all AI-generated text (recall).

	Threshold=0.3	Threshold=0.5	Threshold=0.7	Threshold=0.9
Precision	0.9527	0.9876	0.9973	0.9996
Recall	0.9665	0.9425	0.9188	0.8856
F1	0.9595	0.9645	0.9565	0.9391

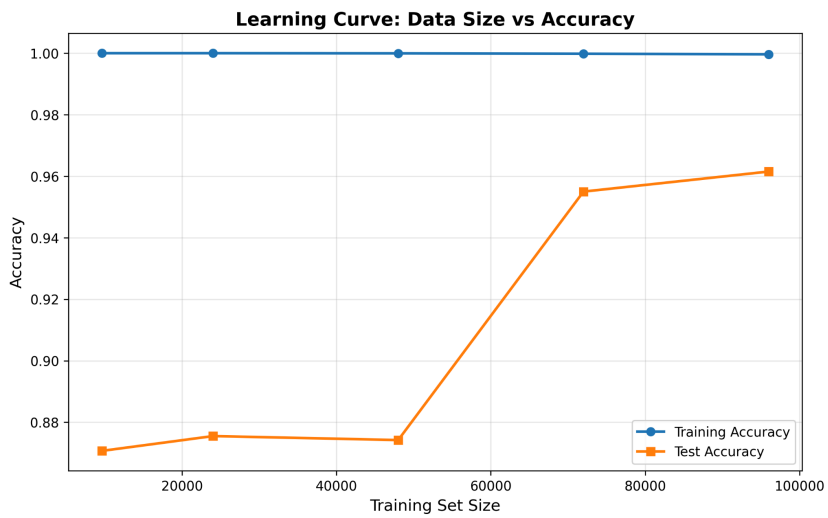


This tradeoff reflects competing priorities. High precision (threshold=0.9) minimizes false positives — critical in contexts where falsely predicting AI usage may have consequences (eg. content moderation). However, this comes at the cost of not detecting 11.4% of AI-generated content. Conversely, high recall (threshold=0.3) ensures comprehensive AI detection but increases false predictions by 4.7%, potentially harming innocent users.

Our selected default threshold of 0.5 represents a pragmatic compromise, balancing user protection (low false positives) while effectively detecting AI-generated text.

2. Training Data Size vs. Accuracy

Training Samples	Train Accuracy	Test Accuracy
9600	1.0000	0.8707
24000	1.0000	0.8755
48000	0.9999	0.8742
72000	0.9998	0.9550
96000	0.9996	0.9615



The model exhibits a non-linear leaning path with a critical inflection point at 72,000 samples, or 75% of data size. Below this threshold, the model struggles to generalize — with 87.4% accuracy at 50% of data size — suggesting insufficient coverage of the variance in stylometric patterns. However, marginal returns diminish sharply beyond the 75% point: the final 25% of data size yields only a +0.65% improvement in accuracy. This plateau suggests the model has generalized to the learnable patterns in our feature space, and additional data simply provides redundant examples rather than new information.

3. Model Complexity vs. Speed

	Logistic Regression	Random Forest (10)	Random Forest (100)	SVM
F1	0.9026	0.9518	0.9552	0.9021
Time (sec)	5.34	5.32	46.36	8.45

While the Random Forest (100 trees) model achieves the best performance (F1=0.9552), it requires 8.7x more training time than the lighter-weight Random Forest (10 trees) or Logistic Regression model. Logistic Regression and SVM show nearly identical performance and fast speeds, suggesting that linear decision boundaries best capture discriminative patterns.

Failure Analysis

We conducted failure analysis on our Random Forest, examining 4,177 misclassified instances (3.48% error rate). Analysis revealed stark platform differences and a critical pattern: 83.9% of failures are False Negatives, or AI-generated text being classified as human, predominantly on Twitter — the platform accounting for 82.9% of all failures.
Here we identified two categories of failures: short human-authored text (insufficient feature information) and short AI-generated text in casual tone (edge case).

1. Example 1: False Positive - Short Human-authored Text

   Platform: Reddit | True Label: Human | Predicted: AI
   Text: “the colts are interviewing trestman for head coach…”

   This post provided insufficient text for reliable feature extraction. The model could not calculate sentiment burstiness for one single sentence — subjectivity and polarity variance both collapse to 0.000000, because variance is mathematically undefined for single data points. The text's ambiguous tone lacks emotional markers, creating a feature profile indistinguishable from AI-generated content. The model's maximum confidence (1.000) shows it treats zero variance as a definitive AI signature rather than genuine stylistic consistency. To improve this in the future, we could analyze alternative micro-features such as capitalization patterns, punctuation density, and lexical uniqueness that remain valid for single sentences, rather than relying on sentence-level variance that requires multiple observations.

2. Example 2: False Negative - Short AI-generated Text in Casual Tone

   Platform: Twitter | True Label: AI | Predicted: Human
   Text: “pls friendship maintenance needs a user manual…”

   This 7-word AI-generated text successfully evaded detection through the use of informal markers. The extremely short output incorporates casual abbreviations ("pls") that usually signal human authorship. The model's zero confidence (0.000) indicates it strongly believed this was human writing, while the degenerate variance features (both 0.0000 due to single-sentence length) provided no discriminative signal. To improve this in the future, we could augment training data with AI text specifically prompted for ultra-short, informal outputs, exposing the model to these evasion patterns. Rather than treating abbreviations as universal human signals, develop context analysis examining whether abbreviations follow authentic usage patterns or appear as isolated tokens.

3. Example 3: False Positive - Short Human-authored Text

   Platform: Reddit | True Label: Human | Predicted: AI
   Text: “boners bbq sometimes these things pretty much write themselves...”

   This 9-word post demonstrates appropriate model uncertainty. The borderline confidence (51%—essentially a coin flip) reveals the classifier recognized the input's ambiguity. The model cannot leverage its primary discriminative features (sentiment variance,

sentence-level patterns), forcing it to make a binary decision on insufficient evidence. To improve this in the future, we could implement multi-stage classification, routing borderline cases to specialized short-text classifiers or flagging them for human review.

Discussion

There are a few findings that surprised us during the evaluation of this project. First, that the feature space is linearly separable. Seeing the marginal difference between linear models like Logistic Regression and SVM compared to Random Forest showed that the detection of AI-generated text in social media contexts did not require complex modeling. Second, that subjectivity and polarity variance accounted for the primary importance of the model — given the ability of AI generators to provide text in various tones, one would think these would be less important features. This suggests that even while using different tones, AI-generated text is more consistent than human-authored. Finally, that removal of punctuation features during the ablation analysis had such a large impact on the model. While it does not seem out of place for AI-generated text to have more statistically consistent punctuation, the size of the impact was a surprising result. It seems there is more to explore considering the conflict between the feature importance and ablation analyses.

We encountered a couple of challenges, namely initial feature engineering and the use of the RoBERTa transformer. Our initial plan was to utilize features through unsupervised learning methods, but a lack of distinct boundaries made this difficult, so we pivoted to a supervised feature engineering approach, manually designing and validating discriminative stylometric, sentiment, and punctuation-based features. For the RoBERTa transformer, we attempted to run this and have a more contextual base for our model. While we attempted to tune hyperparameters to improve efficiency, unfortunately the lack of computational resources was a challenge we were unable to overcome.

If we had additional time and resources, we would not only use the RoBERTa transformer but also focus on a short-text submodel to focus on micro-features and ensure the model is able to detect AI or human-generated text at any sentence length — especially with the importance of variance features to our current model. This could be interpreted as irregularity becoming a proxy for human-authored text authenticity.

For the unsupervised part, the interesting insight was that the AI group showed stylistic complexity but emotional variance, and the human group showed negativity. This may be interpreted as humans tend to show negativity in the context more than human but consistent, AI tends to show emotional variance with complex wordings.

There was a challenge where the initial base model that used TF-IDF did not perform well. After acknowledging imbalance of initial dataset and resolving such issues with teammates, the model with TF-IDF did not perform well. After discussing with teammates, SBERT was applied. Due to its context based understanding(instead of bag-of-words method like tf-idf) and dense vector, silhouette score improved and cross-tabulation result improved significantly.

With more time and resources, insight from LDA, which is that a specific topic was focused by AI(90.3%), could be utilized for improving modeling by adding stylistic features more specifically to AI’s topic.

Ethical Considerations

Our current model focuses on minimizing false positives, meaning there is AI-generated text that is flagged as human-authored. If this solution was deployed in applications for purposes such as disinformation or bot detection, this could have a negative impact as the AI-generated text would appear to be true opinions or grassroot opinions by human users, which could lead to people being swayed ideologically, influencing political discourse. This could potentially be mitigated by lowering the classification threshold or penalizing false negatives more heavily.

Statement of Work

Li-Yuan worked on cleaning and preparation of the Reddit dataset. She led the majority of supervised learning with Logistic Regression and Random Forest and conducted model valuation and analysis including Feature Importance and Ablation Analysis, Sensitivity Analysis, trade-off identification, and Failure Analysis. She also contributed to the final report.

Jessica worked on cleaning and preparation of the Twitter datasets, as well as the AI-generated text dataset creation with the large language model. She worked on supervised learning with SVM and RoBERTa, as well as contributed significantly to the final report.

Sung Jin was responsible for the unsupervised learning models once data cleaning and preparation was complete. He employed three unsupervised learning techniques as well as evaluating, visualizing, and writing the resulting insights.

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